

00 Start Here

00. Start Here and Document Governance

Document purpose

This document is the master source of truth for Midas Learning Cloud. It combines the product vision, platform behavior, user experiences, feature rules, business model, brand direction, legal and compliance considerations, technology recommendations, operating procedures, delivery roadmap, metrics, open questions, and decision history in one controlled reference.

It is deliberately broader than a Product Requirements Document. Individual PRDs, technical specifications, design briefs, legal agreements, operating procedures, and sales materials should inherit their governing decisions from this blueprint.

Intended audience

- Midas Studio founders and business owners.
- Product managers, designers, content creators, and education specialists.
- Ent Hmue and future technical reviewers or implementation partners.
- Engineers responsible for frontend, backend, data, security, and DevOps.
- School owners, school administrators, teachers, and pilot partners.
- Legal, privacy, compliance, and information-security advisers.
- Commercial, marketing, sales, customer-success, and support teams.

How to use this document

1. Read the Executive and Strategy section to understand the business and product direction.
2. Use the Product Ecosystem and platform-specific sections to define user experiences and scope.
3. Use the Feature Modularity section before designing or developing any significant feature.
4. Use the Technology and Data sections as recommendations pending technical review.
5. Use the Legal section as a risk and requirements register, not as legal advice.
6. Use the Operations section to design repeatable school onboarding and support procedures.
7. Record unresolved matters in Open Questions rather than hiding assumptions inside requirements.
8. Record every material change in the Versioning and Change Log.

Authority and decision labels

Every material statement should be understood using the following labels:

- Confirmed: Accepted by the business owner and currently authoritative.
- Recommended: Proposed direction that should be followed unless a better option is approved.
- Assumption: Working condition used to maintain progress; it requires validation.
- TBD: A necessary decision or fact that is not yet known.

- Future: Intentionally excluded from the current release but retained as a possible extension.
- Out of scope: Explicitly excluded from the referenced release or workstream.

Document control rules

- The Markdown file is the canonical editable source.
- The Google Docs version is the collaborative reading and review copy.
- Material decisions must be reflected in both versions.
- Changes affecting multiple sections must be updated consistently across those sections.
- Product decisions must not be silently replaced by implementation choices.
- Technical recommendations remain recommendations until reviewed by Ent Hmue or another approved technical owner.
- Legal and compliance requirements must be reviewed by qualified advisers in the jurisdictions where the platform operates.
- Old decisions should remain visible in the change log when they materially affected prior work.

Master table of contents

1. Start Here and Document Governance
2. Executive Summary and Strategy
3. Product Ecosystem and Core Engines
4. Student Learning Platform
5. Teacher Monitoring Portal
6. Admin Panel and Configuration
7. Feature Modularity and Business Rules
8. Design, UX, Brand and Theme System
9. Marketing, Positioning and Go-to-Market
10. Commercial Model and Revenue Strategy
11. Legal, Privacy, Compliance and Risk
12. Technology Stack and Architecture
13. Data Model and Database Schema
14. Operations, SOPs and Service Management
15. Roadmap, Metrics and Release Readiness
16. Open Questions by Category
17. Appendices and Glossary
18. Versioning and Change Log

01 Executive & Strategy

01. Executive Summary and Strategy

Product definition

Midas Learning Cloud is a multi-tenant, white-label-ready education SaaS platform operated by Midas Studio. It connects three role-based experiences through one shared platform foundation:

1. A gamified Student Learning Platform.
2. A Teacher Monitoring Portal.
3. An Admin Panel for platform and school management.

Students and teachers share one login entry point. The system identifies the authenticated user's role and directs the user to the correct experience. Administrative users use a separate protected admin access point.

Mission

- Help students learn more effectively in a gamified environment, making learning fun and rewarding.
- Help teachers monitor, assess, support, and manage their responsibilities while becoming more involved in student learning and growth.
- Help schools turn curriculum into engaging, personalized, and measurable learning experiences without requiring their own technology team.

Vision

Become a modular education operating platform that allows schools and education organizations to launch branded digital learning environments, configure curriculum and features, measure learning progress, and support students at scale.

Strategic position

Midas Learning Cloud is not intended to be only a quiz application, only a game, only a traditional learning management system, or a full school administration system. Its strategic position is the combination of:

- Engaging student learning.
- Teacher-guided personalization.
- School-controlled curriculum.
- Measurable progress and mastery.
- Configurable gamification.
- Multi-tenant SaaS infrastructure.
- White-label branding and feature control.

Primary customer and users

Category	Primary party	Value received
Paying customer	Schools and education organizations	Branded digital curriculum, measurable learning, reduced technology burden
Daily end user	Students	Engaging, personalized, rewarding learning
Daily operational user	Teachers	Monitoring, assessment, assignment, intervention and support tools
Administrative user	School administrators	Users, classes, content, configuration, branding and reporting
Platform operator	Midas Studio	SaaS revenue, reusable platform IP, curriculum services and partner growth
Future user	Parents or guardians	Visibility and support for student learning

Initial market approach

The initial route is B2B and B2B2C. Midas Studio sells or pilots the platform with schools; schools provide access to teachers and students. A future direct-to-consumer freemium or premium product may reuse the same core learning, content, assessment, progress, and gamification engines.

Initial validation context

- One friendly pilot school.
- Approximately 5-10 students and 1-2 teachers.
- One primary-age mathematics experience.
- Initial learner age approximately 7-8 years.
- Initial topics: addition, subtraction, multiplication, and division.
- Mobile-first responsive web experience.
- Future support for tablets, desktop, iOS, and Android.
- Clickable prototype first, followed by a working pilot MVP.

Strategic priorities

Priority order:

1. Learning outcomes.
2. Student engagement.
3. Teacher usefulness and convenience.

4. School adoption and operational fit.
5. Visual polish.
6. Revenue, paying-school count, and sustainable unit economics.

Business outcomes

The validated product should demonstrate:

- Measurable student learning progress.
- Consistent student participation and completion.
- Teacher ability to identify and support learning needs.
- School willingness to adopt and pay.
- Growth in paying schools and recurring revenue.
- A support model that does not require proportional manual work for every new school.
- Reusable content, configuration, and white-label capabilities.

Guiding strategic constraints

- Do not attempt to build every education feature in the first release.
- Avoid one-off school implementations that cannot become reusable modules.
- Do not allow gamification to obscure learning outcomes.
- Do not collect student data without a defined purpose and access rule.
- Do not make schools dependent on technical knowledge for routine operation.
- Do not make the product architecture dependent on one school, grade, subject, or brand.

02 Product Ecosystem

02. Product Ecosystem and Core Engines

Platform model

Midas Learning Cloud is one SaaS platform with multiple role-based surfaces. The surfaces share authentication, tenancy, permissions, content, learning records, reporting, configuration, and audit services.

Midas Learning Cloud



Tenant hierarchy

Recommended hierarchy:

Platform → Organization → School → Grade → Class → User

- An organization may manage one or more schools.
- A school owns or is assigned its users, curriculum, classes, branding, settings, reports, and learning records.
- A user may have one or more roles, but every action is evaluated within an active tenant and permission context.
- Platform administrators have controlled cross-tenant access for support and operations.

Core learning flow

Login → Onboarding → Screening Assessment → Topic Evaluation
→ Recommended Learning Path → Teacher Review and Approval
→ Lessons, Activities and Missions → Feedback and Retry
→ Progress and Mastery Update → Rewards and Companion Growth

Core engines

Identity and Access Engine

Authenticates users, determines tenant and role context, applies permissions, manages sessions, and directs users to the correct experience.

Curriculum and Content Engine

Stores and organizes Midas content and school-provided content through subjects, curricula, units, topics, lessons, activities, questions, answers, feedback, difficulty, learning objectives, and version states.

Assessment Engine

Presents screening and ongoing assessments, records attempts, evaluates topic-level capability, and produces structured evidence for recommendations and teacher review.

Recommendation and Learning Path Engine

Generates grade-based, subject-based, topic-based, or hybrid learning recommendations. Recommendations may require teacher approval and may respect school rules for student autonomy.

Progress and Mastery Engine

Tracks completion, accuracy, retries, repeated practice, assessment evidence, teacher assessment, and topic mastery. Speed may be a secondary signal but must not become the sole measure of learning.

Assignment and Mission Engine

Allows teachers and authorized users to assign lessons, quizzes, tasks, homework, and missions to individuals, groups, classes, grades, or subjects.

Gamification Engine

Processes points, badges, streaks, missions, achievements, unlockable items, leaderboards, and other motivation mechanics according to school configuration and learner maturity.

Companion Engine

Connects positive learning behaviors to a virtual companion's development. The initial experience is nurturing and positive; higher-age or advanced consequence models require deliberate configuration and review.

Notification Engine

Creates role-appropriate, actionable notifications for assignments, approvals, inactivity, difficulty, achievements, reminders, content changes, and administrative events.

Reporting and Analytics Engine

Aggregates learning, engagement, teacher, content, school, operational, and commercial data into role-appropriate dashboards and reports.

Theme Customization Engine

Applies tenant and school branding, age-appropriate presentation, typography, colors, visual density, illustrations, and future custom-domain or branded-app settings. Detailed requirements will be provided separately.

Subscription and Entitlement Engine

Future engine that maps plans and agreements to feature access, usage limits, premium content, support levels, trial status, billing status, and renewal rules.

Product principles

- Learning outcomes before entertainment.
- Student autonomy within school-controlled boundaries.
- Teacher oversight for consequential recommendations.
- Topic-specific understanding rather than one universal student level.
- Positive motivation by default.
- Age-appropriate UI, feedback, and gamification.
- Configuration before customization code.
- Tenant isolation by design.
- Modular features with explicit ownership and boundaries.
- Accessible and understandable interactions.
- Privacy, safety, and auditability as platform foundations.

Scope layers

Scope layer	Purpose	Minimum result
Clickable prototype	Validate concept, flow and school pitch	Complete representative journey across all three platforms
Pilot MVP	Validate real learning, teacher value and adoption	Real users, content, progress, permissions and basic reporting
Production SaaS	Operate multiple paying schools	Multi-tenancy, security, support, analytics, subscriptions and reliability

Scope layer	Purpose	Minimum result
Expansion platform	Extend subjects, markets and channels	Mobile apps, parent tools, integrations, advanced personalization and white-labeling

03 Student Platform

03. Student Learning Platform

Purpose

Provide students with an age-appropriate, motivating learning environment that makes progress understandable and connects rewards to meaningful learning actions.

Primary information architecture

1. Home / My Learning.
2. Activities, Missions and Leaderboard.
3. Companion.
4. Profile and Progress.
5. Notifications.

Quizzes, tasks, assignments, and homework live within Activities and Missions rather than requiring a separate top-level destination in the initial design.

Shared login and student access

Capability

- Students and teachers use one shared login entry point.
- The system identifies role and tenant after authentication.
- Younger students may use a school-generated username, code, or approved PIN-supported flow.
- Student access must not expose administrative or teacher functions.

Business logic

- Every login resolves an active school and role context.
- Suspended, expired, or unassigned users receive a clear recovery message.
- A student associated with multiple contexts must enter the correct authorized context without seeing data from another tenant.

Admin configuration

- Allowed login methods.
- Password and recovery policy.
- Student code format and expiry.
- Session duration.
- School identity displayed at login.

Onboarding tutorial

Capability

Introduce navigation, learning paths, missions, rewards, companion behavior, progress, and safe-use expectations through a visual guided experience.

Rules

- Younger students receive minimal text and strong visual demonstration.
- The tutorial must be replayable.
- Skipping may be controlled by school or age group.
- Completion is recorded separately from learning progress.

Screening assessment

Capability

Assess addition, subtraction, multiplication, and division across controlled difficulty bands. Produce topic-specific indicators rather than only one universal level.

Inputs

- Student identity and school context.
- Grade or subject context.
- Question topic, difficulty, expected answer, and scoring rules.
- Student response, retry, time, and completion behavior.

Outputs

- Topic-level capability indicators.
- Confidence or evidence level.
- Suggested starting activities.
- Recommended path awaiting teacher approval when configured.

Rules

- Assessment must be short and age appropriate.
- The result must not present a diagnostic or legally significant claim.
- Teachers may approve, reject, or modify recommendations.
- Schools may configure grade-based, subject-based, or hybrid placement.

Home / My Learning

Capability

Give the student an immediate understanding of current work, next recommended action, assigned work, progress, rewards, and companion status.

Required elements

- Continue learning action.
- Recommended next lesson or activity.
- Active assignments and missions.
- Current streak and points where enabled.
- Companion status.
- Recent achievement or encouraging feedback.
- Notification indicator.

UX rule

The student must always understand where they are, what they can do next, how they are progressing, and why the next activity matters.

Learning path

Modes

- Grade-based path.
- Subject-based path.
- Topic-based path.
- Hybrid path combining grade structure with topic flexibility.

Control model

- The school defines the available path modes.
- The system recommends a path.
- The teacher approves or adjusts when approval is required.
- Student exploration outside the recommended path is controlled by school settings.

Lessons and activities

Supported initial forms:

- Explanation or worked example.
- Multiple-choice question.
- Numeric response.
- True or false.
- Matching or sorting.
- Short practice sequence.
- Simple mathematics mini-game.
- Quiz.
- Teacher-assigned task or homework.

Each activity should define its objective, topic, difficulty, expected response, feedback, mastery contribution, reward contribution, age suitability, and version.

Feedback, hints and retry

Younger learners

- Immediate encouraging feedback.
- Short visual or simple-language explanation.
- Optional hint.
- Retry or guided practice.
- No harsh punishment for incorrect answers.

Older or higher-education learners

- More concise and academically mature feedback.
- Configurable retry limits.
- More explicit scoring and consequence rules.
- Reduced character emphasis where appropriate.
- Any negative or loss mechanic requires age, pedagogy, and school review.

Activities, missions, assignments and homework

- Activities are individual learning interactions.
- Missions are goal-oriented collections of actions.
- Assignments are teacher-directed requirements.
- Homework is an assignment intended for completion outside class.
- A teacher can assign to an individual, group, class, grade, or subject cohort.
- Due dates, completion criteria, availability, and late behavior are configurable.
- Completion updates student progress and teacher reporting.

Gamification

Initial mechanics:

- Points.
- Badges.
- Streaks.
- Missions.
- Achievement milestones.
- Avatar.
- Optional leaderboard.
- Companion growth.
- Future cosmetic unlocks.

Rewards must be connected to meaningful learning actions. The product must not reward random tapping, repeated low-value actions, or behavior that undermines learning.

Companion

The companion is a nurturing motivational layer inspired by virtual-pet experiences.

Initial capabilities:

- Companion identity and visual state.
- Growth points earned through learning.
- Positive reactions to progress.
- Simple care or encouragement actions.
- Unlockable visual progress.

Initial restrictions:

- No default death, abandonment, severe decay, or anxiety-inducing punishment.
- No monetized pressure aimed at children.
- No dependency on daily perfect attendance.

Leaderboard

- Disabled by default until enabled by the school.
- Configurable by class, cohort, subject, or event.
- Identity display may use real name, nickname, avatar, or anonymous rank.
- Profanity and unsafe nickname controls are mandatory.
- Schools may choose alternative cooperative or personal-best rankings.
- Historical records remain protected when the feature is hidden.

Profile and progress

Students may view:

- Completed topics and lessons.
- Topic mastery indicators.
- Current assignments and missions.
- Points, badges, streaks, and achievements.
- Companion development.
- Teacher-approved learning direction.
- Age-appropriate explanation of progress.

Notifications

Student notifications may include:

- New assignment or mission.
- Upcoming due item.
- Teacher-approved learning path.
- Achievement earned.
- Companion milestone.
- Safe reminder to continue learning.

Notifications should be actionable, rate-limited, age appropriate, and configurable by school policy.

Student success criteria

- Students can enter and understand the product with minimal adult instruction.
- The next meaningful action is always clear.

- Activities provide immediate and useful feedback.
- Learning progress is visible without shaming comparison.
- Gamification improves participation without replacing learning.
- The platform works effectively on mobile and tablet browsers.

04 Teacher Portal

04. Teacher Monitoring Portal

Purpose

Help teachers quickly understand class and student progress, identify learning needs, approve or adjust recommendations, assign work, record professional judgment, and take action without excessive administrative complexity.

Primary navigation

- Dashboard.
- Classes.
- Students.
- Learning Paths.
- Homework and Assignments.
- Notifications.
- Reports.

Teacher dashboard

The dashboard should prioritize actionable information:

- Active and inactive learners.
- Assignment completion.
- Topic mastery and learning gaps.
- Pending learning-path approvals.
- Students needing attention.
- Recent achievements.
- Upcoming or overdue work.

The dashboard should avoid presenting data without a clear interpretation or next action.

Class management

Teachers can:

- View assigned classes and cohorts.
- View class membership.
- Filter by grade, subject, topic, activity, or status.
- Review class-level progress.
- Assign content to the class.
- Identify groups for intervention or extension.

Teachers cannot view classes outside their authorized scope.

Student monitoring

Teacher views should include:

- Student profile and class context.

- Screening and assessment history.
- Topic-level performance.
- Completed and incomplete work.
- Accuracy, retries, and practice history.
- Mastery indicators.
- Recent engagement.
- Teacher notes and interventions.
- Recommended and approved learning path.

Recommendation approval

Workflow

1. System creates a recommendation.
2. Teacher receives a pending approval notification.
3. Teacher reviews topic evidence and proposed path.
4. Teacher approves, modifies, rejects, or defers.
5. Decision is recorded with timestamp, actor, and rationale where required.
6. Student experience updates according to the approved decision.

Rules

- The platform must not represent an unapproved recommendation as a teacher decision.
- Schools can configure when approval is mandatory.
- Teacher adjustment must not remove historical assessment evidence.
- Recommendation history should remain auditable.

Homework and assignments

Teachers can:

- Select approved lessons, activities, quizzes, or missions.
- Assign to students, groups, classes, grades, or subjects.
- Define availability, due date, completion threshold, and retry behavior.
- Monitor started, completed, overdue, and excused states.
- Provide feedback or follow-up action.

Teacher assessment and notes

- Teachers can record observations, intervention notes, and assessment judgments.
- Notes have visibility rules: private teacher, authorized staff, school record, or future parent-visible.
- Sensitive notes require stricter permissions and audit history.
- Teacher evidence may contribute to mastery only through defined rules.

Notifications and alerts

Alerts should be prioritized by actionability:

- Pending recommendation approval.
- Repeated difficulty.
- Significant inactivity.
- Incomplete or overdue work.
- Strong improvement or milestone.
- Content or assignment problem.
- Administrative message.

Teachers should be able to configure delivery frequency within school policy.

Reporting

Initial reports:

- Student progress summary.
- Class progress summary.
- Topic mastery summary.
- Assignment completion.
- Assessment results.
- Engagement and activity summary.

Reports should clearly state whether a measure is product activity, inferred mastery, or teacher-provided judgment.

Future teacher gamification

Teacher gamification is future scope. If introduced, it should encourage constructive actions such as timely feedback, intervention completion, or professional collaboration. It must not encourage superficial competition or distort teacher priorities.

Teacher success criteria

- A teacher understands class status quickly.
- Students needing attention are easy to identify.
- Recommendations can be reviewed with adequate evidence.
- Assignments can be created without complex setup.
- Monitoring reduces uncertainty rather than adding reporting burden.
- Teacher permissions remain limited to authorized students and classes.

05 Admin & Configuration

05. Admin Panel and Configuration

Purpose

Provide Midas Studio and authorized school administrators with controlled tools to manage tenants, users, curriculum, features, branding, analytics, subscriptions, safety, and platform operation.

Administrative role separation

Platform Owner / Super Admin

- Global platform configuration.
- Tenant creation and suspension.
- Platform-wide feature availability.
- Global content and template management.
- Cross-tenant support access.
- Subscription and commercial administration.
- Security and audit oversight.

Platform Operations Admin

- School onboarding.
- Content support.
- Configuration assistance.
- User support.
- Operational reporting.

School Admin

- School users and classes.
- School content and assignments.
- School branding and themes within entitlement.
- School feature visibility.
- School reporting.
- Teacher access and responsibilities.

Future specialist roles

- Curriculum Manager.
- Content Reviewer.
- Billing Administrator.
- Support Agent.
- Data or Reporting Viewer.

Admin navigation

- Overview.
- Organizations and Schools.
- Users and Roles.
- Grades, Classes and Subjects.
- Curriculum and Content.
- Assessment and Learning Configuration.
- Gamification and Companion.
- Branding and Themes.
- Feature Visibility and Entitlements.
- Analytics and Reports.
- Subscription and Billing.
- Safety, Audit and Platform Settings.

School and tenant management

Capabilities:

- Create and configure organization and school tenants.
- Assign school identifier, status, plan, branding, locale, and time zone.
- Configure supported grades, subjects, classes, and terms.
- Suspend or archive a tenant without immediate destructive deletion.
- Export or delete tenant data according to approved policy.

User and role management

- Invite, create, import, deactivate, reactivate, and remove users.
- Assign one or more roles within scoped contexts.
- Assign teachers to classes and subjects.
- Assign students to classes, grades, and learning contexts.
- Review failed access and account status.
- Apply bulk actions with confirmation and audit history.

Curriculum and content management

Content lifecycle

Draft → Review → Approved → Published → Revised / Archived

Required capabilities

- Create subjects, curricula, units, topics, lessons, and activities.
- Upload or reference school-provided material.
- Create questions, answers, explanations, hints, and feedback.
- Tag content by age, grade, subject, topic, difficulty, objective, and ownership.
- Preview student presentation.
- Review and approve content.

- Publish to selected tenant, grade, class, or cohort.
- Version content without silently changing historical evidence.
- Archive or retire content.

Ownership labels

- Midas Original.
- School Owned.
- School Licensed.
- Third-Party Licensed.
- Co-created.

Assessment and learning configuration

Configurable items:

- Assessment topics.
- Difficulty bands.
- Question selection rules.
- Scoring and confidence thresholds.
- Recommendation logic.
- Teacher approval requirements.
- Grade-based, subject-based, topic-based, or hybrid path mode.
- Student autonomy.
- Mastery thresholds.
- Retry and feedback behavior by education level.

Gamification configuration

- Points and earning rules.
- Badge definitions.
- Streak behavior.
- Mission templates.
- Reward visibility.
- Companion growth rules.
- Leaderboard enablement and identity mode.
- Age-based restrictions.
- School-specific overrides within allowed boundaries.

Branding and theme configuration

Initial controls:

- School name.
- Logo.
- Primary and supporting colors.
- Welcome message.
- Basic visual theme.

Future Theme Customization Engine requirements will cover typography, illustration, component style, visual density, age mode, character system, custom domains, and branded mobile experiences.

Feature visibility and entitlements

Every significant non-core feature must support configuration by one or more of:

- Platform.
- Subscription plan.
- Organization.
- School.
- Role.
- Grade.
- Class.
- Subject.
- User group.

Disabling a feature should remove access without corrupting historical data. Re-enabling should restore appropriate prior state unless a documented deletion policy applies.

Safety and moderation controls

- Profanity filtering.
- Unsafe nickname restriction.
- Content report handling.
- Content publication approval.
- User suspension.
- Audit logs.
- Rate limiting and abuse controls.
- Age-appropriate defaults.

Analytics and reporting

Admin reporting may cover:

- School activation and usage.
- Student and teacher activity.
- Learning and mastery indicators.
- Assignment completion.
- Content performance.
- Feature adoption.
- Operational support volume.
- Subscription usage and entitlement.
- Commercial metrics for Midas administrators.

Admin success criteria

- Routine school configuration does not require code changes.

- Tenant boundaries are clear and enforced.
- Content has controlled review and versioning.
- Every major administrative action is attributable.
- Feature visibility is predictable and recoverable.
- Administrators can understand the impact before publishing or disabling changes.

06 Feature Modularity

06. Feature Modularity and Business Rules

Modular product requirement

Every significant feature outside core platform foundations must be designed as a module. A module must be understandable, testable, configurable, and replaceable without requiring unrelated parts of the product to be rewritten.

Core foundations versus modular features

Centralized core foundations

- Identity and authentication.
- Tenant and organization context.
- User roles and permissions.
- Shared configuration resolution.
- Audit history.
- Shared event and notification infrastructure.
- Common analytics conventions.
- Data protection and security controls.

Modular product features

- Content management.
- Assessment.
- Recommendation and learning paths.
- Lessons and activities.
- Homework and assignments.
- Progress and mastery.
- Missions.
- Rewards and badges.
- Streaks.
- Companion.
- Leaderboards.
- Notifications.
- Reports.
- Theme customization.
- Subscription and billing.
- Parent experience.

Mandatory module specification

Every significant feature specification must include:

Area	Required documentation
Purpose	User problem, business value, intended outcome
Users	Roles, contexts and eligibility
UX	Entry points, screens, states, actions and feedback
Inputs	Data, configuration, user actions and events consumed
Logic	Business rules, calculations, decisions and state transitions
Outputs	UI states, records, events, notifications and reports produced
Frontend	Components, client validation, responsive and accessibility behavior
Backend	Services, commands, queries, jobs and transaction boundaries
Data	Entities, fields, relationships, retention and ownership
API	Endpoints or contracts, authorization, requests, responses and errors
Configuration	Defaults, override scope, validation and administrative owner
Permissions	Read, create, update, approve, publish, disable and delete rights
Visibility	Feature flags, entitlements and tenant rules
Analytics	Events, measures and reporting expectations
Safety	Abuse cases, moderation, privacy and sensitive-data considerations
Edge cases	Failure, conflict, offline, duplication, stale state and recovery
Testing	Unit, integration, permission, end-to-end and regression coverage
Operations	Monitoring, support, rollback and incident ownership
Evolution	Future extension points and migration considerations

General business rules

Access

- Access is denied unless explicitly allowed by role and active tenant context.
- Users see only schools, classes, content, records, and features within scope.
- Multiple roles may exist on one account, but context must be explicit.
- Privileged actions require stronger controls and audit records.

Learning model

- Schools and teachers can use grade-based, subject-based, topic-based, or hybrid paths according to permission.
- A student may have different capability levels across subjects and topics.
- Teacher professional judgment can modify system recommendations.
- Historical evidence remains available after recommendation changes.

Feedback

- Younger students receive encouraging, visual, low-text feedback.
- Older students may receive more mature scoring, retry, and consequence rules.
- Feedback configuration must not create humiliating or harmful experiences.

Content

- Only authorized, approved content may be published.
- Content ownership and license status must be recorded.
- Published content changes require versioning.
- Student history must reference the content version used at the time.

Configuration hierarchy

Recommended precedence:

Platform safety rule

→ Subscription entitlement

→ Organization configuration

→ School configuration

→ Grade / class / subject configuration

→ Authorized teacher assignment

→ Student-specific learning state

Lower levels cannot override mandatory platform safety or legal controls.

Feature disablement

- Hide the feature from affected interfaces.
- Prevent new feature actions.
- Retain historical records unless deletion is approved.
- Preserve referential integrity for reports and audit.
- Provide clear behavior for re-enablement.

Notifications

- Notifications must have a purpose, recipient, trigger, priority, and action.
- Duplicate or excessive notifications must be prevented.
- Sensitive information should not appear in insecure channels.

Auditability

Record significant actions such as role changes, content publication, recommendation approval, feature configuration, tenant status changes, data export, and deletion.

Definition of feature completion

A significant feature is not complete because the screen exists. It is complete for a release only when the approved scope includes working UX, business logic, backend behavior, data handling, permissions, configuration, analytics, errors, tests, documentation, monitoring, and operational ownership.

07 UX, Brand & Themes

07. Design, UX, Brand and Theme System

Experience strategy

The platform uses one shared design foundation with role- and age-appropriate expressions.

Surface: Younger student | Intended character: Friendly, colorful, cartoon-inspired, visual and motivational

Surface: Older student | Intended character: More mature, focused, independent and academically appropriate

Surface: Teacher | Intended character: Calm, clear, supportive, data-informed and action-oriented

Surface: Admin | Intended character: Structured, precise, configurable and professional

Age-based evolution

The product should grow with the learner. Age or education-level configuration may alter:

- Typography.
- Illustration and character prominence.
- Color intensity.
- Visual density.
- Language complexity.
- Navigation guidance.
- Feedback style.
- Gamification intensity.
- Retry and scoring rules.
- Companion visibility.

The underlying learning records and core engines remain consistent even when presentation changes.

Student UX principles

- Prefer visual explanation over text for younger students.
- Maintain a single obvious primary action per screen.
- Use large, forgiving touch targets.
- Show progress and purpose.
- Provide immediate feedback.
- Avoid shame, fear, or manipulative pressure.
- Make errors recoverable.
- Keep navigation shallow and predictable.

Teacher UX principles

- Lead with decisions and actions rather than raw data.
- Surface students needing attention.
- Explain why recommendations were generated.

- Reduce repetitive setup.
- Preserve filters and context when navigating.
- Keep sensitive information appropriately restricted.

Admin UX principles

- Separate platform-level and tenant-level controls.
- Show configuration scope and impact.
- Require confirmation for high-impact actions.
- Support draft, preview, publish, rollback, and audit where appropriate.
- Avoid hidden inheritance rules.
- Make feature and entitlement state understandable.

Responsive strategy

- Web application first.
- Mobile-first student experience.
- Tablet support as a primary learning context.
- Responsive desktop support for all portals.
- Teacher and admin workflows may use denser desktop layouts while retaining essential mobile usability.
- Native iOS and Android applications are future channels using the same backend and rules.

Accessibility requirements

- Readable text and scalable typography.
- Sufficient contrast.
- Keyboard navigation for web workflows.
- Visible focus states.
- Screen-reader-compatible labels and structure.
- No meaning conveyed by color alone.
- Reduced-motion support.
- Clear error descriptions and recovery actions.
- Touch targets suitable for children and mobile use.
- Plain language appropriate to the audience.
- Alternative content for meaningful images and media.

Shared design system

The design system should define:

- Semantic color tokens.
- Typography roles by audience and age mode.
- Spacing and layout grid.
- Buttons and interactive states.
- Forms and validation.

- Cards and surfaces.
- Navigation patterns.
- Tables, progress indicators, badges, alerts, and notifications.
- Icons, illustrations, avatar and character rules.
- Motion and reduced-motion behavior.
- Responsive breakpoints.
- Accessibility acceptance criteria.

White-label model

White-labeling should be configuration-driven. The same codebase should support different school identities without creating separate forks.

Initial controls:

- School name.
- Logo.
- Primary color.
- Supporting colors.
- Welcome message.

Future Theme Customization Engine considerations:

- Typography packs.
- Illustration style.
- Component radius and surface style.
- Character or mascot pack.
- Age-mode presets.
- Custom domain.
- Branded email and notification templates.
- Branded native application configuration.
- Theme preview, validation, versioning, and rollback.

Detailed theme-engine requirements remain TBD and will be supplied later.

08 Marketing & GTM

08. Marketing, Positioning and Go-to-Market

Brand architecture

- Midas Studio: company or studio identity responsible for creating and operating valuable products.
- Midas Learning Cloud: recommended working name for the education SaaS platform.
- School-branded experience: tenant-specific identity shown to students and teachers according to white-label configuration.

Meaning of Midas

The Midas concept represents creating lasting value: what the studio builds should become valuable for the immediate customer and reusable for future users and partners.

Positioning statement

For schools that want engaging digital learning without building their own technology team, Midas Learning Cloud is a configurable education SaaS platform that combines gamified student learning, teacher monitoring, school-controlled curriculum, and measurable progress in one white-label-ready system.

Value propositions

Student value

Learning feels clear, interactive, rewarding, and responsive to individual progress.

Teacher value

Teachers gain practical visibility and tools to support learning without navigating an overly complex system.

School value

Schools can digitize and brand curriculum, manage access and features, and measure learning without developing their own platform.

Midas Studio value

Midas builds reusable product IP, recurring SaaS revenue, curriculum-service revenue, and a platform capable of supporting multiple partners.

Initial go-to-market

1. Create a credible demonstration.
2. Present it to the friendly pilot school.
3. Agree on a controlled pilot.

4. Measure learning, engagement, teacher usefulness, and operational effort.
5. Convert the pilot into a paid plan or obtain a clear reason for non-conversion.
6. Convert evidence into a case study and repeatable sales narrative.

School pitch narrative

The pitch should focus on outcomes rather than features:

We turn your curriculum into an engaging digital learning experience with personalized student activities, teacher monitoring, school-controlled configuration, and measurable progress - without requiring your school to build a technology team.

Marketing assets eventually required

- Product website.
- School pitch deck.
- Product demo.
- Pilot proposal.
- Case study.
- Pricing overview.
- Security and privacy overview.
- Feature catalogue.
- Implementation and onboarding guide.
- Teacher and administrator quick-start materials.

Segmentation considerations

Potential future segments:

- Private primary schools.
- School groups.
- Tutoring centers.
- Education agencies.
- Homeschool programs.
- Curriculum publishers.
- Training institutions.

The first segment should remain narrow until product-market evidence exists.

09 Commercial Model

09. Commercial Model and Revenue Strategy

Commercial direction

The initial model is B2B SaaS supported by implementation and curriculum services. A future consumer model may be added without weakening the school product.

Revenue components

Setup fee

May cover tenant creation, configuration, initial branding, user preparation, content setup, training, and launch support.

Recurring subscription

May be monthly or annual and may be based on plan, school size, active students, classes, or negotiated enterprise terms.

Per-student component

Useful when platform value and support cost scale with active learners. Minimum commitments may be required to avoid uneconomic small accounts.

Curriculum conversion and customization

School curriculum conversion, custom activity design, specialist content work, and bespoke reports may be included at a defined level or priced separately.

Premium content and features

Future premium tiers may unlock additional curricula, subjects, grades, analytics, gamification, integrations, support, and branding capabilities.

Recommended packaging direction

Package: Pilot | Intended customer: First validation partner | Illustrative scope: Limited students, one subject, guided support, fixed duration

Package: Starter | Intended customer: Small school or limited rollout | Illustrative scope: Core learning, teacher portal, basic branding, standard support

Package: Professional | Intended customer: Growing school | Illustrative scope: More students, richer reporting, configuration and content options

Package: Enterprise | Intended customer: School group or strategic partner | Illustrative scope: Multiple schools, advanced controls, integrations, service commitments

Final plan names, prices, limits, and entitlements are TBD.

Pilot commercial rules

- Free or discounted access must have a defined duration.
- Scope, students, teachers, content, support, and responsibilities must be written.
- The intended paid model should be discussed before the pilot begins.
- Pilot feedback and measurement access are part of the exchange.
- Conversion, extension, or termination should occur at the end of the agreed period.

Commercial success metrics

- Number of pilot schools.
- Pilot-to-paid conversion rate.
- Number of paying schools.
- Active students under contract.
- Setup revenue.
- Monthly and annual recurring revenue.
- Average revenue per school.
- Gross margin and support cost per school.
- Retention and churn.
- Expansion revenue.
- Time to onboard a school.

Pricing principles

- Price for value and service burden, not only feature count.
- Keep the first offer understandable.
- Separate repeatable platform value from bespoke services.
- Avoid unlimited customization inside a standard subscription.
- Make premium and enterprise entitlements explicit.
- Review pricing after real pilot evidence.

Future consumer model

A freemium consumer experience may provide limited content and levels. Premium may unlock more subjects, grades, activities, learning paths, reports, and cosmetic or companion features. Consumer monetization involving children requires careful ethical, legal, and parental review.

10 Legal & Compliance

10. Legal, Privacy, Compliance and Risk

Important limitation

This section documents business requirements and risk areas. It is not legal advice. Qualified legal counsel must confirm agreements and obligations in each operating jurisdiction.

Platform and school responsibilities

Recommended operating position:

The school is responsible for the legality, accuracy, educational suitability, and permissions associated with content it supplies. Midas Studio provides the technology platform and processes the content according to agreed instructions.

This position does not remove all platform responsibility. Midas remains responsible for its own processing, security, product behavior, original content, representations, and applicable legal duties.

Curriculum and intellectual property

Requirements:

- Record content owner and license status.
- Require schools to confirm rights to supplied content.
- Obtain a license to host, reproduce, transform, and display content as needed for service delivery.
- Distinguish school content from Midas original content.
- Define ownership of co-created and AI-assisted materials.
- Provide a content complaint and takedown process.
- Avoid using school content for unrelated customers without authorization.

Student and child data

- Collect only data required for defined product purposes.
- Define legal basis and consent requirements.
- Define school, parent, and platform roles.
- Provide age-appropriate notices where required.
- Restrict use of student data for unrelated advertising or profiling.
- Define retention, export, correction, and deletion procedures.
- Review cross-border storage or processing.

Privacy documentation

Likely future documents:

- Privacy policy.
- School data-processing agreement.
- Parent or guardian notice and consent where required.

- Data retention schedule.
- Subprocessor list.
- Data subject request procedure.
- Data breach response procedure.

Security obligations

- Encryption in transit.
- Secure password hashing.
- Role and tenant isolation.
- Least-privilege access.
- Input validation and safe file handling.
- Audit logging.
- Backup and recovery.
- Vulnerability and dependency management.
- Incident response.
- Security review before real student use.

Child safety and content safety

- Profanity filtering and nickname controls.
- Age-appropriate content review.
- Reporting mechanism.
- Moderation and escalation procedure.
- Restrictions on social interaction.
- Careful treatment of leaderboards, streaks, and loss mechanics.
- No manipulative monetization directed at children.

Agreements likely required

- Pilot agreement.
- SaaS subscription agreement.
- Data-processing agreement.
- Content license or curriculum services schedule.
- Implementation or customization statement of work.
- Acceptable-use terms.
- Privacy notices and consent materials.
- Subcontractor and vendor agreements.

Liability allocation concepts

- School warranty regarding supplied content rights and suitability.
- Midas warranty regarding authorized platform service.
- Defined service limitations.
- Mutual confidentiality.
- Indemnity concepts subject to counsel.

- Limitation of liability subject to counsel.
- Takedown and suspension rights.
- Data incident responsibilities.

Compliance areas to investigate

- Child privacy laws.
- General data-protection law.
- Student-record law.
- Consumer protection.
- Accessibility.
- Copyright and licensing.
- Online safety.
- Payment and subscription requirements.
- Education-specific regulation.
- Data residency and cross-border transfer.

Risk register summary

Risk	Example impact	Initial mitigation
Unauthorized curriculum	Copyright claim	Content warranty, license records, takedown process
Child-data misuse	Regulatory and trust damage	Data minimization, scoped use, consent and legal review
Tenant data exposure	Severe privacy incident	Tenant-scoped authorization and permission testing
Harmful gamification	Student anxiety or reputational harm	Positive defaults, school controls, age review
Inaccurate learning claims	Misleading product representation	Clearly label indicators and involve teachers
Excessive customization	Unsustainable delivery cost	Modular configuration and priced services
Security incident	Service and legal impact	Security controls, monitoring, response and backups

11 Technology & Architecture

11. Technology Stack and Architecture

Status of technical recommendations

The stack below is recommended for review and confirmation by Ent Hmue. The architectural requirements are more important than any single vendor or framework choice.

Technical principles

- Modular and readable.
- Secure by default.
- Multi-tenant from the foundation.
- Mobile-first and API-ready.
- Documented and transferable.
- Testable at business-rule and permission boundaries.
- Managed infrastructure where it reduces operational burden.
- Capable of future iOS and Android clients.

Recommended stack

Layer	Recommendation	Rationale
Web frontend	Next.js with TypeScript	Responsive app routing, shared types and maintainable component architecture
UI system	Reusable component library and design tokens	Consistency, accessibility and white-label theming
Backend	Modular TypeScript API layer	Shared language, clear modules and typed contracts
Database	PostgreSQL	Strong relational model for tenancy, users, curriculum and progress
Authentication	Managed or well-supported identity system	Secure sessions, recovery and role integration
File storage	Managed object storage	Curriculum files, images, audio and media assets
Hosting	Managed cloud platform	Easier deployment, scaling, observability and rollback

Layer	Recommendation	Rationale
Source control	Git repository with protected workflow	Traceability and collaboration
Testing	Unit, integration, authorization and end-to-end	Protect business rules and critical journeys
Monitoring	Logs, metrics, errors and security signals	Reliable operations and incident detection

Recommended architecture style

Start with a modular monolith: one deployable application or tightly coordinated application set with strong internal module boundaries. This is simpler to build and operate than premature microservices while preserving future extraction paths.

Logical architecture

Web Clients

- └─ Student Experience
- └─ Teacher Portal
- └─ Admin Panel

|

API and Application Layer

- └─ Identity and Tenant Context
- └─ User and Role Module
- └─ Content Module
- └─ Assessment Module
- └─ Learning Path Module
- └─ Progress and Mastery Module
- └─ Assignment Module
- └─ Gamification Module
- └─ Companion Module
- └─ Notification Module
- └─ Reporting Module
- └─ Theme Module
- └─ Subscription Module
- └─ Audit Module

|

Data and Infrastructure

- └─ PostgreSQL
- └─ Object Storage
- └─ Cache / Queue when justified
- └─ Monitoring and Logs
- └─ Backup and Recovery

Frontend architecture

- Shared design tokens and accessible primitives.
- Separate route groups or applications for student, teacher, and admin experiences.
- Feature folders aligned with backend modules.
- Typed API client.
- Server and client state responsibilities documented.
- Responsive behavior and age-mode theming.
- Error boundaries, loading states, and offline messaging.

Backend architecture

- Modules own their business logic and data access boundaries.
- Controllers or route handlers should not contain core business logic.
- Cross-module changes use explicit services or domain events.
- Background work uses queues or scheduled jobs only when needed.
- Authorization occurs on the server for every protected action.
- Configuration resolution is centralized and auditable.

API principles

- Versioned contracts where breaking change risk exists.
- Typed requests and responses.
- Consistent error format.
- Tenant context verified server-side.
- Idempotency for retry-sensitive commands.
- Pagination for large collections.
- Rate limits for exposed or abuse-prone operations.
- API documentation maintained with implementation.

Security architecture

- Secure session and token management.
- Password hashing through proven libraries or managed identity.
- Server-side permission enforcement.
- Tenant-scoped database access patterns.
- Input validation.
- Secure file upload checks.
- Secrets stored outside source code.
- Dependency and vulnerability review.
- Audit logs protected from ordinary alteration.

DevOps

- Separate development, staging, and production environments.
- Automated build and deployment checks.

- Reviewed database migrations.
- Environment-variable management.
- Backup and restore procedures.
- Deployment rollback.
- Uptime and error monitoring.
- Incident and post-incident review process.

Testing strategy

- Unit tests for scoring, mastery, recommendation and reward rules.
- Integration tests for database and module collaboration.
- Authorization tests for role and tenant isolation.
- End-to-end tests for student, teacher and admin critical paths.
- Migration tests for data changes.
- Accessibility checks for major screens.
- Performance tests before large school rollout.

Technical handover requirements for Ent Hmue

- Confirm or amend the recommended stack.
- Review modular-monolith boundaries.
- Review tenancy and authorization design.
- Validate database schema and migration approach.
- Validate authentication and account-recovery design.
- Define hosting, environment and deployment strategy.
- Define coding standards and repository structure.
- Define API and event conventions.
- Identify security risks and technical unknowns.
- Produce an implementation sequence and effort estimate.

12 Data & Schema

12. Data Model and Database Schema

Data design principles

- Every tenant-owned record has an explicit tenant or school relationship.
- Ownership and visibility are distinct concepts.
- Historical learning evidence should remain interpretable after content changes.
- Sensitive data is minimized and classified.
- Records use stable identifiers.
- Creation, update, actor, and version metadata exist where relevant.
- Destructive deletion is controlled and auditable.

Core entity groups

Tenancy and organization

- Organization.
- School.
- School Settings.
- Subscription.
- Feature Entitlement.
- Theme.
- Theme Version.

Identity and access

- User.
- User Profile.
- Role.
- Permission.
- User Role Assignment.
- Session.
- Invitation.
- Class Membership.

Academic structure

- Grade.
- Class.
- Academic Term.
- Subject.
- Curriculum Program.
- Unit.
- Topic.
- Learning Objective.

Content

- Lesson.
- Activity.
- Question.
- Answer Option.
- Correct Response Rule.
- Hint.
- Explanation.
- Content Version.
- Content Ownership Record.
- Content Publication.
- Media Asset.

Assessment and learning

- Assessment.
- Assessment Item.
- Assessment Attempt.
- Student Response.
- Topic Evaluation.
- Recommendation.
- Recommendation Decision.
- Learning Path.
- Learning Path Item.
- Progress Record.
- Mastery Record.
- Teacher Assessment.
- Teacher Note.

Assignment and engagement

- Assignment.
- Assignment Target.
- Assignment Attempt.
- Mission.
- Mission Progress.
- Points Ledger.
- Badge.
- Student Badge.
- Streak.
- Companion.
- Companion State.
- Reward.
- Leaderboard Configuration.
- Leaderboard Entry or Derived Ranking.

Communication and operations

- Notification.
- Notification Preference.
- Report Definition.
- Report Export.
- Support Request.
- Audit Log.
- Data Export Request.
- Data Deletion Request.

Key relationship overview

Organization 1-* School

School 1-* Grade / Class / UserRole / Theme / ContentPublication

User *-* Role through UserRoleAssignment

Teacher *-* Class through scoped assignment

Student *-* Class through ClassMembership

Subject 1-* Topic 1-* Lesson 1-* Activity

Activity 1-* Question 1-* AnswerOption

Student 1-* AssessmentAttempt 1-* StudentResponse

Student 1-* ProgressRecord / MasteryRecord / Recommendation

Teacher 1-* RecommendationDecision / Assignment / TeacherNote

Student 1-1 Companion and 1-* PointsLedger / Badge / MissionProgress

Tenant isolation requirement

Every query and command involving tenant data must validate active tenant context. A school identifier supplied by the client is not sufficient authorization. Server-side access checks must confirm role, tenant, resource, and action.

Content versioning requirement

- Student responses reference the exact content version presented.
- Publishing a revision creates a new version rather than silently rewriting history.
- Reports can explain which version produced historical results.
- Archived versions remain available to authorized reporting and audit processes.

Progress and mastery distinction

- Progress describes completion and position within assigned or available learning.
- Mastery describes evidence of understanding against defined criteria.
- Completion alone does not equal mastery.
- Teacher assessment may supplement system evidence.
- Mastery logic must record the evidence and rule version used.

Points ledger principle

Points should use a ledger rather than only a mutable total. Each change records reason, source event, value, timestamp, and reversal where necessary. The displayed total is derived from valid ledger entries.

Audit log principle

Audit records should capture actor, tenant, action, target, result, relevant change summary, and time. Sensitive values should not be copied into logs unnecessarily.

Data retention

Retention periods are TBD and must be defined by data category, purpose, contract, legal obligation, and school offboarding policy.

13 Operations & SOPs

13. Operations, SOPs and Service Management

School onboarding SOP

Stage 1: Qualification

- Confirm school goals, users, curriculum, technical readiness, timeline, and decision maker.
- Confirm whether the engagement is demonstration, pilot, or paid rollout.
- Identify legal and data requirements.
- Record requested customization separately from standard configuration.

Stage 2: Agreement and readiness

- Execute appropriate pilot or commercial agreement.
- Confirm content rights and data responsibilities.
- Define success measures and review schedule.
- Confirm support contacts and escalation paths.

Stage 3: Tenant setup

- Create organization and school tenant.
- Configure identity, branding, grades, classes, subjects, features, and access methods.
- Create administrator and teacher accounts.
- Validate tenant isolation and permissions.

Stage 4: Content setup

- Receive approved source material.
- Record ownership and license status.
- Convert content into the platform structure.
- Review learning objectives, difficulty, answers, feedback, and age suitability.
- Obtain authorized approval before publication.

Stage 5: User setup and training

- Create or import users.
- Assign roles and classes.
- Train administrators and teachers.
- Provide student onboarding instructions.
- Complete launch-readiness checklist.

Stage 6: Launch and support

- Activate the tenant.

- Monitor access, errors, and adoption.
- Hold scheduled check-ins.
- Record issues and decisions.

Stage 7: Review and conversion

- Review learning, engagement, teacher, operational, and commercial evidence.
- Decide whether to convert, extend, modify, or stop.
- Document the reason and next action.

Content production SOP

1. Intake and ownership check.
2. Curriculum mapping.
3. Lesson and activity design.
4. Question, answer, hint, and feedback creation.
5. Age and difficulty review.
6. Educational review.
7. Technical validation.
8. Student preview.
9. Authorized approval.
10. Publication.
11. Performance monitoring.
12. Revision or retirement.

AI-assisted content must remain subject to human review before student publication.

Configuration change SOP

- Identify requested change and affected scope.
- Confirm requester authority.
- Review impact on users, data, reporting, and entitlements.
- Preview where supported.
- Apply with audit record.
- Verify the affected experience.
- Roll back or escalate if incorrect.

Feature rollout SOP

- Define intended users and outcome.
- Confirm module readiness.
- Enable in development and staging.
- Validate permissions, configuration and analytics.
- Pilot with limited tenant or cohort.
- Monitor errors and behavior.
- Expand gradually.
- Document release and rollback conditions.

Incident management SOP

1. Detect and record.
2. Classify severity and affected tenants.
3. Contain risk.
4. Communicate to authorized parties.
5. Restore service safely.
6. Preserve evidence and logs.
7. Conduct root-cause review.
8. Implement prevention and update documentation.

Support model

Support categories:

- Access and account issues.
- Teacher and student usability.
- Content errors.
- Configuration requests.
- Data and report questions.
- Platform defects.
- Security or privacy concerns.
- Commercial and subscription questions.

Each category needs owner, priority, target response, escalation route, and closure criteria.

School offboarding SOP

- Confirm authorized termination request.
- Disable new access according to contract.
- Export permitted school data.
- Resolve outstanding invoices and support matters.
- Apply retention and deletion policy.
- Remove or archive branding and content access.
- Record completion and responsible actor.

14 Roadmap & Metrics

14. Roadmap, Metrics and Release Readiness

Delivery phases

Phase 1: Concept and blueprint

- Consolidated product and business definition.
- Clickable experience direction.
- Technical review.
- Pilot-school alignment.

Phase 2: Clickable prototype

- Representative Student Platform.
- Teacher Monitoring Portal.
- Admin configuration experience.
- Fictional school and demo data.
- One end-to-end demonstration flow.

Phase 3: Pilot MVP

- Real authentication and roles.
- One tenant and controlled real users.
- Mathematics content.
- Assessment and recommendation.
- Teacher approval.
- Activities, progress, rewards and reporting.
- Basic admin content and configuration.

Phase 4: Production SaaS

- Hardened multi-tenancy.
- Security, monitoring, backup and support.
- Repeatable school onboarding.
- Content versioning and operational controls.
- Subscription and entitlement foundations.

Phase 5: Expansion

- More schools, subjects and grades.
- Parent experience.
- Advanced analytics.
- Native iOS and Android.
- Advanced white-labeling.
- Integrations and AI-assisted workflows.

Core metrics

Learning

- Assessment completion.
- Topic-level improvement.
- Mastery evidence.
- Quiz accuracy and retry behavior.
- Homework completion.
- Teacher assessment of progress.

Student engagement

- Activation.
- Weekly active students.
- Lesson starts and completions.
- Return frequency.
- Mission completion.
- Companion interaction.
- Streak participation where enabled.

Teacher value

- Teacher activation.
- Dashboard usage.
- Recommendation review time.
- Assignment creation and completion.
- Intervention actions.
- Teacher satisfaction and reported effort.

School adoption

- Pilot completion.
- Pilot-to-paid conversion.
- Active schools.
- Paying schools.
- Retention and renewal.
- Feature adoption.
- Support volume and onboarding time.

Commercial

- Setup revenue.
- Monthly and annual recurring revenue.
- Average revenue per school.
- Gross margin.
- Cost to onboard and support.
- Expansion revenue.

- Churn.

Quality and reliability

- Error rate.
- Availability.
- Response time.
- Failed login and permission errors.
- Content defect rate.
- Support resolution time.
- Security and privacy incidents.

Prototype release criteria

- Main journey works: login, onboarding, assessment, recommendation, teacher approval, activity, reward, progress, monitoring and admin configuration.
- Student, teacher and admin navigation is understandable.
- Phone and tablet layouts are usable.
- No broken critical interactions.
- Representative mathematics activities exist.
- School branding and feature controls are visible.
- The product can be presented confidently in a short school demonstration.

Pilot MVP release criteria

- Real authentication and role routing.
- Tenant-scoped access.
- Approved content and student accounts.
- Assessment and recommendation records.
- Teacher approval workflow.
- Activity completion and progress persistence.
- Basic reporting.
- Backup, monitoring and support plan.
- Privacy, pilot agreement and content rights reviewed.
- Critical permission and end-to-end tests pass.

Production readiness criteria

- Multi-tenant isolation verified.
- Security review completed.
- Data retention and deletion defined.
- Operational monitoring and incident response active.
- Repeatable school onboarding documented.
- Subscription and entitlement behavior defined.
- Performance and reliability meet agreed targets.
- Legal and commercial documents approved.
- Technical and operational ownership assigned.

Go / no-go principle

A release should not proceed because a date has arrived. It proceeds when the agreed learning, product, technical, security, legal, operational, and commercial criteria for that phase are satisfied or explicitly accepted as documented risks.

15 Open Questions

15. Open Questions by Category

Open-question management

Each open question should have a category, question, recommended direction, decision owner, priority, target decision point, status, and impact if unresolved.

Product

- What final product name should replace or confirm Midas Learning Cloud?
- Which capabilities are mandatory for the first working MVP?
- Which capabilities are explicitly deferred?
- How much student autonomy is default versus school configured?
- What is the first post-mathematics subject?

Student experience

- What final age and grade context will the pilot use?
- What character or creature should the companion be?
- What companion interactions are appropriate?
- How should UI maturity change across age bands?
- Which cooperative alternatives to leaderboards should be included?

Teacher experience

- What evidence must teachers see before approving a recommendation?
- How much can teachers override school defaults?
- Which alerts are essential for the pilot?
- Should teachers create content directly in the first production release?
- Which reports are required by the pilot school?

Admin and configuration

- Which settings belong to Midas Studio versus school administrators?
- Which specialist admin roles are needed first?
- What bulk import formats are required?
- What configuration inheritance and override model will be adopted?
- What audit visibility should schools receive?

Content and curriculum

- Which curriculum standard will the first pilot use?
- Who provides final educational approval?
- What are the first 10-20 activities?
- What content formats and media are required?
- What is the exact AI-assisted content-review workflow?

- How are co-created content rights handled?

Assessment and mastery

- What scoring model produces topic recommendations?
- How is confidence represented?
- What evidence and threshold define mastery?
- How does teacher assessment affect mastery?
- How often should reassessment occur?

Gamification

- Which Yu-kai Chou-inspired motivation principles should be explicitly adopted?
- What are the initial point and badge rules?
- How are streaks protected from harmful pressure?
- What reward economy, if any, should exist?
- What mechanics are appropriate for older learners?

Branding and theme customization

- What exact controls will the Theme Customization Engine provide?
- Can schools upload fonts, illustrations and character packs?
- Which theme changes require Midas approval?
- How are themes previewed, versioned and rolled back?
- When are custom domains and branded apps offered?

Marketing and strategy

- Which school segment should be targeted first?
- What competitor set should be evaluated?
- What evidence is required for the first case study?
- What claims can be made responsibly?
- What is the initial sales process and owner?

Commercial

- What is the pilot duration?
- What conversion offer follows the pilot?
- What setup fee is viable?
- Is recurring pricing per school, student, plan, or hybrid?
- What customization is included versus separately charged?
- What school-count and revenue targets define validation?

Legal and compliance

- Which countries and jurisdictions apply?
- What consent and notice model is required?
- What data-controller and processor roles apply?

- What retention schedule is required?
- What insurance and contract terms are appropriate?
- What accessibility obligations apply?

Technology

- Which stack choices will Ent Hmue confirm or amend?
- Which authentication provider should be used?
- Which hosting platform and region should be selected?
- How should tenant isolation be implemented and tested?
- When are queue, cache, search, or separate services justified?
- What offline or low-connectivity behavior is required?

Data and analytics

- Which student fields are essential?
- What events may be used for product analytics?
- What data can be aggregated across tenants?
- Which reports require export?
- How long is each data category retained?

Operations

- Who owns school onboarding?
- Who converts and reviews curriculum?
- What support commitments can Midas sustain?
- What is the incident escalation tree?
- How are school data exports and deletion requests fulfilled?

16 Appendices & Glossary

16. Appendices and Glossary

Appendix A: Glossary

Term	Meaning
SaaS	Software delivered as an ongoing service, normally through subscription
Multi-tenant	One platform serving multiple organizations while keeping their data and configuration isolated
Tenant	An isolated customer environment, normally an organization or school
White-label	Configuring the platform to appear under a partner or school identity
Authentication	Proving who a user is
Authorization	Deciding what an authenticated user may access or do
Role-based access control	Permissions assigned through roles such as teacher or school admin
Feature flag	A controlled switch that enables or disables a product capability
Entitlement	A commercial or administrative rule granting access to a feature or limit
Curriculum	Structured educational content and learning objectives
Learning path	Ordered or recommended learning activities for a student
Assessment	A structured activity used to collect evidence of capability or progress
Mastery	Evidence that a learner understands a defined topic or objective
Progress	Completion and position within learning activities or assignments

Term	Meaning
Gamification	Use of game-inspired mechanics to support motivation and engagement
Mission	A goal-oriented set of learning actions
Companion	A virtual character that develops through learning activity
Modular monolith	One deployable system with strong internal feature boundaries
API	A documented software interface used by modules or clients to exchange data and actions
Frontend	The user-facing application interface
Backend	Server-side services, rules, integrations and data access
Database schema	The structure and relationships used to store data
DevOps	Practices and systems for building, deploying, monitoring and operating software
CI/CD	Automated integration, testing and deployment workflow
Audit log	Protected record of important actions and changes
MVP	Minimum Viable Product: the smallest working release that tests core value
MLP	Minimum Lovable Product: a small release designed to be useful and emotionally engaging
Pilot	Controlled real-world trial with defined scope and success criteria
B2B	Business-to-business commercial model
B2B2C	A business sells through another organization to reach end users
B2C	Business-to-consumer commercial model
Freemium	Basic free access with paid upgrades

Term	Meaning
Recurring revenue	Revenue received repeatedly through subscription or contract

Appendix B: Role and access summary

Role	Scope	Core capabilities
Platform Owner	Global	Platform, tenant, security, commercial and feature control
Platform Operations Admin	Authorized cross-tenant	Onboarding, support, content and configuration operations
School Admin	One school or authorized group	Users, classes, content, branding, features and reports
Teacher	Assigned school, classes and subjects	Monitor, approve, assign, assess and report
Student	Own learning context	Learn, complete activities, receive feedback and view progress
Parent / Guardian	Future scoped child relationship	View and support approved student progress

Appendix C: Reusable feature specification template

1. Feature name and purpose.
2. User problem and business outcome.
3. Roles and eligibility.
4. Entry points and user journey.
5. UI states and interaction behavior.
6. Inputs.
7. Business rules and state transitions.
8. Outputs and events.
9. Frontend requirements.
10. Backend requirements.
11. Data entities and retention.
12. API contracts.
13. Admin configuration.
14. Permissions and visibility.
15. Analytics and reporting.

16. Notifications.
17. Safety, privacy and abuse considerations.
18. Edge cases and error recovery.
19. Acceptance criteria.
20. Test coverage.
21. Monitoring and support.
22. Future extension points.

Appendix D: Decision record template

Field	Description
Decision	Clear statement of what was decided
Context	Problem and constraints
Options	Alternatives considered
Outcome	Selected direction
Reason	Why it was selected
Owner	Person accountable for the decision
Impact	Product, business, legal, technical or operational effects
Revisit condition	Evidence or event that would reopen the decision

Appendix E: Technical review checklist for Ent Hmue

- Validate requirements and release boundaries.
- Confirm frontend and backend approach.
- Confirm authentication and tenant model.
- Review permission enforcement.
- Review module boundaries.
- Review database entities and relationships.
- Review content and progress versioning.
- Confirm API conventions.
- Confirm hosting, DevOps, monitoring, backups and recovery.
- Identify security, privacy, performance and scalability risks.
- Produce implementation plan, estimates and technical decision records.

Appendix F: Future supporting documents

- Student Platform PRD.
- Teacher Portal PRD.

- Admin Panel PRD.
- Theme Customization Engine specification.
- Assessment and Recommendation specification.
- Data and analytics specification.
- Security and privacy architecture.
- School onboarding SOP.
- Content production SOP.
- Pilot agreement and measurement plan.
- Commercial packaging and pricing document.

17 Versioning Log

17. Versioning and Change Log

Versioning policy

- Major version: fundamental business, platform, architecture, or operating-model change.
- Minor version: meaningful section, feature, policy, or scope addition.
- Patch version: clarification, correction, formatting, or non-material improvement.
- Every change records the reason, owner, affected sections, and status.
- Superseded decisions remain traceable when they materially influenced prior work.

Change log

Version	Date	Change Summary	Reason	Affected Sections	Owner	Status
0.1	TBD	Initial consolidated Midas Learning Cloud blueprint	Establish one master source of truth	All	Midas Studio	Working

Pending version milestones

Target Version	Trigger
0.2	Business owner review and corrections completed
0.3	Ent Hmue technical review incorporated
0.4	Pilot school requirements and curriculum incorporated
0.5	Legal and privacy review incorporated
1.0	Approved baseline for pilot MVP planning

Change request procedure

1. Describe the requested change and reason.
2. Identify affected sections, modules, users, data, contracts, and roadmap items.
3. Identify decision owner and required reviewers.
4. Evaluate compatibility, migration, cost, risk, and timing.
5. Approve, reject, defer, or request more evidence.
6. Update the canonical Markdown and Google Docs version.

7. Add the version entry and communicate material changes.